

Forecasting Code

Main GOAL

My goal is to build out a forecasting tool for mid to long term stocks for my stock portfolio that is working on trading in the background

Investing Ideology

We will look at a large amount of stocks generally

If a stock is sitting in profit for 3 day or more then we add a 5% trailing stop loss

News Checker

What is the new saying before the market will be what will be looked at but then also look at the most changes in a stock to see where shorts are occurring

Data Scrapper

Putting the data into a excel to do the rest of the analysis from fred or other websites to get a consitnaitly set of data that is stored lets decide on how first

Short term data

Stock Price extraction

And what ever else we need for the metrics

Long Term Forecasting 3-4 quarters

1. Additive Decomposition
2. Multiplicative Decomposition
3. Min (Additive, Multiplicative)
4. Max (Additive, Multiplicative)
5. Average (Additive, Multiplicative)
6. Simple Weighted Average (k=4)
7. Weighted 4-Quarter Moving Average ($W=[0.4 \ 0.3 \ 0.2 \ 0.1]$)
8. Simple Exponential Smoothing
9. Holt's linear trend method
10. Holt's-Winter's multiplicative method
11. OLS linear trend + quarterly dummies
12. AutoBest (METHOD 12 Performed optimization from the 11 methods cited above, based on an 8-period holdout, forwards and backwards, optimized for minimal RMSE)

Short-term forecasting

Depending on metric we forecast different amounts, or we will actively monitor it

Not sure on how to acquire this data per stock but being able to analyze would be good

I think there should be a long term forecasting and short term forecasting

Use correlation forecasting to create what is the closest correction in terms for stock metrics like volume 2 3 day volatility

- **Relative Strength Index (RSI):**
 - Measures the speed and change of price movements, identifying overbought (>70) or oversold (<30) conditions.
- **Moving Averages (EMA/SMA):**
 - 50-day and 200-day lines indicate trend direction, with EMA reacting faster to recent price changes, suitable for short-term traders.
- **Short Interest Ratio / Days to Cover:**
 - Calculates shares sold short divided by average daily volume. A ratio over 5-10 indicates high squeeze potential, as shorts struggle to cover.
- **Stochastic Oscillator:**
 - Compares closing prices to range over time, with readings below 20 suggesting oversold and above 80 suggesting overbought, similar to RSI.
- **Bollinger Bands:**
 - Measures volatility. Prices reaching the upper or lower bands often suggest a trend reversal.
- **Trading & Sentiment Metrics**
 - **Volume:**
 - Validates price movements; high volume during a price surge indicates strong buying conviction.
 - **MACD (Moving Average Convergence Divergence):**
 - Shows the relationship between two moving averages, used to spot trend changes.
 - **Price-to-Sales (P/S Ratio):**
 - Evaluates how much investors pay for each dollar of sales, useful for valuing stocks with little to no earnings.
 - **Risk-Reward Ratio:**
 - Evaluates if a trade is worth the potential risk, aiming for higher potential gain than loss.
 - **Slow vs. Fast Momentum:**
 - Long-term trends (e.g., 12 months) are often termed "slow," while "fast" momentum refers to reacting to short-term market shifts.
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Valuer for each metric

Using the data we will decide on long term or shortterm

Investor

This is will actually put in the trade using what we decide

For long term i want it to be decided and show me the top 20 options and the data used to come to the conclusion and we will decide how long we want to stay in the stock this happens before the market opens

During the time when the market it open the For short term if there is a strong correlation and we can geniuailly see a strong upside then we will invest and keep an eye on it